

Syeifa Andhika Nugroho

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Profile

"D4 student in Electronic Engineering Technology at Politeknik Negeri Semarang with a strong foundation in hardware design and a particular interest in Internet of Things (IoT) technology. Has practical experience in assembling electronic systems, troubleshooting circuits, and integrating microcontrollers with smart sensors. Enthusiastic about creating connected technology solutions and ready to contribute to solving industry challenges.

Experience

EMBEDDED SYSTEMS & ELECTRONICS DEVELOPER | FREELANCE | 2025

- Design and prototype small-scale electronic systems and custom IoT solutions based on microcontrollers (ESP32, Arduino, etc.) according to client specifications and requirements.
- Integrating various sensors and actuators and programming system control logic using C/C++ in the Arduino IDE or VS Code.
- Performing hardware troubleshooting and periodic system functionality testing to ensure hardware stability and reliability before delivery to the client.
- Actively communicating with clients to interpret technical requirements, manage project timelines, and ensure the final product meets operational targets.

HARDWARE & PCB DESIGNER | FREELANCE | 2025

- Designing circuit schematics (schematic capture) and performing custom multi-layer PCB layout (PCB routing) using EDA software (such as Altium Designer, Eagle, or KiCad).
- Selecting optimal components (component selection) while considering power efficiency, component availability on the market, and the physical dimensions of the board.
- Performing independent assembly through manual soldering of both SMD and THT components on prototype boards.
- Testing circuit continuity and performance using electronic measuring instruments (multimeters, oscilloscopes) to minimize the risk of noise or hardware malfunctions.

Education

BACHELOR OF APPLIED SCIENCE IN ELECTRONIC ENGINEERING | 2023 – PRESENT | STATE POLYTECHNIC OF SEMARANG

Current GPA: 3.60 out of 4.0. Focusing on Embedded Systems and Internet of Things (IoT). Relevant coursework includes Microcontroller Programming, PCB Design, and Industrial Automation. Actively involved in hands-on projects integrating hardware with IoT networks.

HIGH SCHOOL DIPLOMA IN NATURAL SCIENCES | 2020 – 2023 | SMA N 1 DEMAK

Built a strong foundation in analytical thinking, mathematics, and physics. Developed logical problem-solving skills that serve as the core groundwork for understanding electronics engineering and programming concepts.

Skills & Abilities

- Internet of Things
- Embedded Systems
- PCB & Circuit Design
- Microcontroller Programming
- Electronics Troubleshooting
- Project Management
- Teamwork & Collaboration

Engineering & Hardware Projects

Autonomous Legged Robot for Contour and Stair Exploration

Partner: Politeknik Negeri Semarang

Year : 2024

Value : IDR 500,000

Role : Hardware Engineer

Detail : A specialized legged robotics project designed to navigate steep vertical obstacles and uneven terrain, such as stairs.



Smart Irrigation System Prototype

Partner: Politeknik Negeri Semarang

Year : 2025

Value : IDR 200,000

Role : Software Engineer

Detail : An IoT-based smart irrigation system developed using an ESP32 microcontroller to monitor soil moisture levels in real time. This system features automatic irrigation control via a relay-controlled pump using a threshold-based method, as well as data integration with an IoT dashboard for remote monitoring.



SOS Bracelet

Partner : Independent Client

Year : 2025

Value : IDR 200,000 (Freelance Commission)

Role : Hardware Engineer

Detail : IoT wearable safety device. It functions as a panic button system that instantly sends emergency notifications and location data via a BLE network to a smartphone, which then forwards the information to the cloud.



4 DOF Robotic Arm

Partner : Polines Robotics Evolution

Year : 2025

Value : IDR 450,000

Role : Hardware Developer & Software Engineer

Detail : A simple 4-DoF (degrees of freedom) robotic arm project designed for a children's educational module. This robot is controlled via a smartphone.



Autonomus Legged SAR Robot

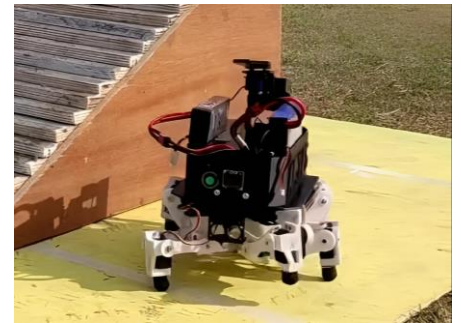
Partner : Independent Project

Year : 2025

Value : IDR 350,000 (Self-funded)

Role : Hardware Modifier & Integrator

Detail : Engineered a custom portable power and connectivity solution by retrofitting a Xiaomi AC1200 Wi-Fi extender with an integrated UPS module. Designed and assembled an internal power bank using 18650 lithium-ion cells, featuring flexible dual-input charging (12V DC & USB-C) and a dedicated 12V output for external routers.



Role and Association

- Member of Polines Robotics Evolution (2025 - 2026)
- Student Member of Politeknik Negeri Semarang Electronic Engineering Community